

NEURALTM
INDUSTRIAL AUTOMATION

Redefining innovation and excellence»

**PROCESS MEASURING AND
CONTROL INSTRUMENTS MANUFACTURER**

www.neuralindustrialautomation.in



We, at Neural Industrial Automation Pvt. Ltd, help in making enterprises safe, smart and sustainable with our automated electronic controllers and instruments. Primarily to meet the requirements of clean room customers, we manufacture and deal in the supply of security, process measurement, control systems and other relevant systems.

Mission Statement

- Provide High Quality, Affordable Systems.
- Create and cultivate long term relationship with clients.
- Respond immediately to the changing needs of clients.
- Achieve complete customer satisfaction.
- Improve our services continuously.

Vision

The vision is to build a highly trained and efficient team who aid in the process of manufacturing of electronic controllers and instruments, and customer service personnel. Our personnel is trained to face anything that may occur and are quick to respond to customer needs and concerns.

At Neural every person in our company is a member of team and a team player, with the expectation and need that they perform their duties to their fullest capacities and potential. Our Objective is a commitment to the highest Quality Electronic Controllers and Customer Service.

Mission

Neural Industrial Automation is a young and dynamic company specialized in manufacture and supply of Door interlocking system and sensors. Below are the products we supply.

- **Door Interlock System**
 - 2 Door to 8 doors interlocking
- **Pass box Interlocking System**
 - Push Button
 - IR based
 - Display Type
 - No Touch
 - Delay Timer
 - Smart Controller
 - FLP Kepad
- **FLP Interlocking System**
 - Pass box
 - Door interlocking
- **Air Shower Controller**
- **LAF/BUFF Controller**
- **Bio Safety**
- **Differential pressure Indicator**

Door Interlocking System:

In clean room areas, any personnel passing through an unclassified area to a classified area is required to pass through a series of rooms. In each room, there are certain steps that need to be followed. This procedure ensures that the person is not carrying any external contamination into the process area and also the pressure in each room is maintained before opening the door. Door Interlock Controllers achieve this by preventing simultaneous access of doors by ensuring that when one door of a room is open then the other door is closed. Push buttons are available for door access. Emergency stations are available for opening of all doors. There is a delay between each door opening to maintain required pressure. Door status can be read by any master system connected over RS485 network. This door interlocking system shall interface with the fire alarm system.

Features:

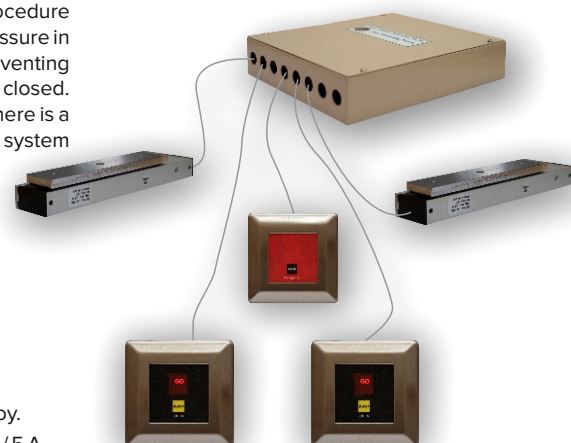
- Micro controller based.
- Supports 2 to 8 doors as per requirement.
- It can support Emergency Stations.
- Delay time between door openings is 4 Sec.
- Inbuilt sensor Magnetic Lock to read door feedback.
- Relay status indication
- Door status indication

Specifications:

- Push Button Inputs : 12 No's
- Door Feedback Inputs : 12 No's
- LED indication Outputs : 12 No's
- Door control outputs : 12 No's

RELAYS:

- Contact material : Silver Alloy.
- Contact Rating : 230V AC / 5 A.
- No. of Operations Mechanical : 10⁷, Electrical : 10⁵



Push button - Soft touch

- Keypad: Soft Touch Membrane Key
- Elegant look North-West SS faceplate.
- Indication facility for Door Entry permitted / prohibited.
- Support an entry switch for opening the Door.
- Long Life BAR LED for door open/close indication.
- Elegant look North-west SS Face plate with appropriate curvature to prevent resting of dust & powder.
- Easy mounting arrangement for fixing on door frames or on Wall
- GO message for Door Entry permitted
- WAIT message for Door Entry prohibited
- Life: 100,000 cycles



Push button - No-touch

- Keypad: No- touch membrane key
- Green LED indication for Door Entry permitted
- Red LED indication for Door Entry prohibited
- Life: 100,000 cycles
- Enclosure type: SS faceplate
- Mounting : Flush mounting



Pass Box Interlocking System

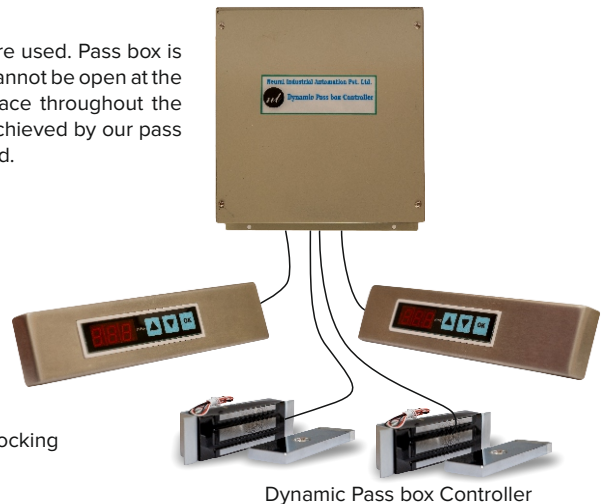
In the process of material transfer from unclassified area to classified area pass boxes are used. Pass box is rectangular box with two doors. The two doors are interlocked such a way that two doors cannot be open at the same time, due to which pressure losses and bacterial contamination doesn't take place throughout the material transferring process. The prevention of simultaneous access of doors is been achieved by our pass box controller by ensuring that when one door for a pass box is open then other doors closed.

Static Pass Box Controller

- Advanced Micro controller based.
- Inbuilt sensor Magnetic Lock to read door feedback.
- Status indication: GO, WAIT, DOOR OPEN, CLOSE DOOR
- UV Lamp ON for both Doors closed /UV Lamp OFF for both Doors Open
- CF Lamp ON when any door is open/CF Lamp OFF for both Doors Closed

Features:

- Advanced micro controller based 2Door or 3 Door STATIC/DYNAMIC Pass box interlocking system.
- Status indications with High intensity RED color BAR LED s.
- Door Status indications like "GO", "WAIT", "CLOSE DOOR", "DOOR OPEN" messages or
- "LOCKED", "RELEASED" for every pushbutton.
- Buzzer output for door is kept open more than pre-programmed time from 1-30 min.
- When Door1 is opened the display on the Door 2 and Door 3 will indicate "WAIT" and at the Door1 the display will indicate "DOOR OPEN".
- When any of the Door is kept open beyond a particular time say 2 min, buzzer will be blown and the display at the doors will indicate "CLOSE DOOR".
- UV Lamp controlling.
- Blower fan controlling.
- CF Lamp controlling.
- All Doors are provided with Electromagnetic locks/Electric Strikers.
- 3 digit 0.39"7-Segment RED LED to display the timer value.
- Soft-touch membrane keypad provided to program timer values.
- 3mm Diffused red LED for Blower failure indications.
- Soft-touch membrane keypad provided to program timer value.
- Soft- touch membrane keypad provided to control LAF/BUF controls.
- 3mm Diffused BI-Color LED indication to display ON/OFF status of UV, CF, Blower.
- Door delay time: pre-programmed as per the requirement.



Specifications:

- Mode of operation: NC or NO.
- Pushbutton inputs based on Doors Max up to 3 Doors.
- Door Feedback inputs based on Doors Max up to 3 Doors.
- Door control Outputs based on Doors Max up to 3 Doors.
- LED indications for push buttons: GO/WAIT.

Relay Specifications:

- Contact material : Silver Alloy.
- Contact Rating : 230V AC / 2A.
- No. Of Operations
- Mechanical : 10⁷.
- Electrical : 10⁵.
- Input voltage : 24 Watt at 80-280VAC, 50Hz.
- Operating voltage : 12V / 24V DC.

Pass Box Interlocking System - Keypad Models



Smart Controller



Push Button



No Touch



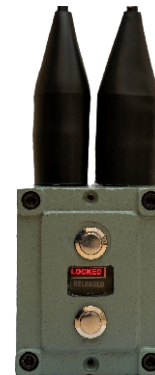
Display Type



Delay Timer



IR Keypad



FLP Keypad

Air Shower Controller

The Air shower is a supporting device for the clean workshops and the assembly clean room, is used to blow the dust attached to the surfaces of the people and objects entering the clean workshop. It also functions as an air brake to prevent non-purified air from entering the clean area.

The air shower system controls the sequential locking and release of the doors, and controls the blower. The two doors are interlocked in such a way that the two doors cannot be opened at the same time which prevent pressure losses and cross contamination.

Our Air Shower interlocking system achieves this by preventing simultaneous access of doors by ensuring that when one door for an Air Shower interlocking system is open, then the other door is closed, before opening of door

1. When any one of the doors is opened, the push button on the other doors will indicate "WAIT" and the person accesses the door by pressing push button
2. All doors are provided with EM Locks



Features

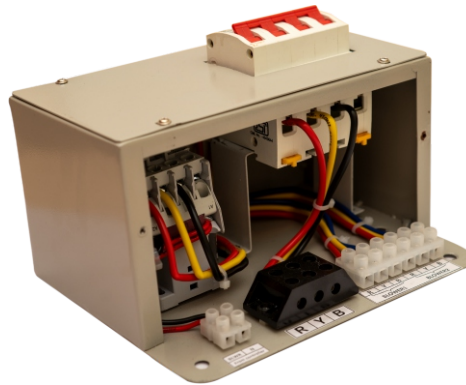
- Advanced micro controller based.
- Supports Maximum of 2 doors.
- Supports Electromagnetic Locks/Electric strikers.
- Door status indication with high intensity Bar LEDs.
- Status indication: GO, WAIT on push button station.
- Digital Display with soft tactile keypad.
- Status indication: OPEN (Exit door open).
- Mode of operation: MAN, AUTO on digital display module.
- User programmable Blower ON time variable from 0-255 Sec.
- Buzzer output for the Exit Door is open.

Pushbutton

- Keypad: Soft Touch Tactile Membrane Key.
- Status LED s: WAIT Message for Door Open / Entry Prohibited.
- GO Message for Door Close / Entry Permitted.
- Audible Indication: Buzzer.
- Life: 100,000 Cycles Min.
- Enclosure type: North West SS faceplate.
- Mounting: Flush mounting.
- Dimension: 90mm L x 90mm W x 25mm D.

Specifications

- Mode of operation: NC or NO.
- Door delay time: pre programmed as per requirement.
- Push button inputs based on Doors maximum 3 doors.
- Door feedback inputs based on Doors maximum 3 doors.
- Door control outputs based on Doors maximum 3 doors.
- LED indications for push buttons: GO/WAIT.
- Input voltage: 24 Watt at 80-280VAC, 50Hz.
- Operating voltage: 12V/24V DC.
- Dimensions for DPM: 110mm L x 90mm W x 20mm D.



LAF/BUF Controller

BUF ON/OFF Controller is used for ON/OFF controlling of Blower, UV Lamp & CF Lamp in Dynamic pass boxes, Bio safety cabinets and laminar air flow. The status of Blower, UV Lamp & CF Lamp is provided on display.

Specifications

- Blower Relay O/P: 230V AC / 2A (up to 1HP Blower can be connected).
- UV Lamp Relay O/P: 230V AC / 2A.
- CF Lamp Relay O/P: 230V AC / 2A.
- No. of blower failure indications: 5.
- Relay Specifications:
- Contact material: Silver Alloy.
- Contact Rating: 230V AC / 2 A.
- No. of Operations
- Mechanical: 10⁷
- Electrical: 10⁵.
- Enclosure: MS Powder Coated for Controller unit SS-316 face plate for DPM.
- Mounting: Panel.
- Power: 100-260V, AC 50 Hz.
- DPM dimensions: 150mm L x 91mm W x 29mm D.

Features

- Advanced Micro controller based.
- ON/OFF Message Display.
- Relay O/P for Blower (up to 1 HP), UV Lamp & CF Lamp.
- ON/OFF Message status indication with high intensity Bar LEDs.
- Status indication: ON, OFF-Blower, ON, OFF-UV Lamp & ON, OFF –CF Lamp.
- Blower Failure Status Indications.
- Soft touch Membrane keypad.
- Easy for Installation.



Temperature, RH and DP Indicators

The NIA "Clean Room Process Monitor" has been specifically developed keeping process measurement requirements of CLEAN ROOMS in Pharmaceutical industries. The Process Monitor can measure Temperature, RH Inbuilt integral Sensor, Differential Pressure Inbuilt Sensor or three analog inputs or two analog inputs and inbuilt differential pressure sensor. It has an on demand dew point display. The monitor has status LEDs for process violation indication. RH, Temperature and Differential pressure will be displayed individually and data transfers to any master device on MODBUS RS-485 communication. It also has a communication diagnosis LED. The monitor has an inbuilt buzzer for audible process violation. It has a keypad to program and calibrate the instrument by user. It has a stainless steel Modular Faceplate, which is designed as a mounting accessory for monitors suited for brick and modular walls for Clean Room applications where you need a flush mounting mechanism.



Features:

- Advanced Microcontroller based process measurement.
- Software programmable channel ranges and units.
- Programmable resolution.
- 4 Digit, 0.39"/0.56"/0.8"/1.0"/1.2"/1.5"/1.8"/2.0"bright 7- Segment LED display for displaying Temperature, RH and DP.
- Process parameter status indication with 3 Bi-Color #mm LEDs.
- User selectable set limits for process value violation.
- Inbuilt buzzer to audible process value violation.
- Password protection for calibration and programming mode.
- Software calibration for 3 channels.
- MODBUS RTU multi drop RS-485 serial communication for PLC, SCADA, etc.
- Programmable baud rate between 9.6k/19.2k/38.4k.
- Diagnosis indication of communication with master.
- Remote programmable from master device.
- User programmable offset correction.
- Brick and modular wall mounting.
- Stainless steel enclosure.

Specifications

Input Type:

- Inbuilt integral Sensors/4-20mA/0 to 10VDC.

Measuring Ranges:

- %RH: 0.0 to 100%RH.
- Temp: -40°C to 100°C.
- DP: -105.4 to 105.4 mm WC, -1034 to 1034 Pascals and 8 types of units with equivalent values.

Accuracy:

- %RH: $\pm 2\%$ RH for 10 to 90% and $\pm 4\%$ RH for remaining.
- Temperature: $\pm 0.3^\circ\text{C}$ for 0°C to 50°C and 2°C for remaining.
- DP: $\pm 1.5\%$ for all units.

Resolution:

- %RH: 1/0.1% RH.
- Temp: 1/0.1°C.
- DP: 0.1 mm WC.

Power supply:

- 24VDC/24VAC.

Digital Differential Pressure Indicator

The NIA "Differential Pressure Indicator" is specially designed to indicate the pressure difference between two different areas. The large 4 digit, 0.56"7 segment red LED display is easy to read, even from a distance. Alarm high and low set limit can be configured, which can help the user to monitor the DP and to take necessary actions to maintain the room conditions as per the requirement or application. Differential Pressure Indicator displays data transfers to any master device on MODBUS RS-485 communication. It has Stainless steel Modular Face plate, which is designed as a mounting accessory for monitor suited for Modular wall Clean Room applications where you need flush mounting mechanism.



Features:

- Advanced Microcontroller based DP measurement.
- Software programmable channel ranges, units & resolution.
- 4 Digit, 0.56"bright LED display for displaying differential pressure.
- Process parameter status indication with LEDs.
- Process value violations display indication.
- User Selectable set limits for process value violation.
- Inbuilt buzzer to audible process value violation.
- Password protection for programming mode.
- Software calibration.

Specifications

- **Input Type** : Inbuilt Sensor/4-20mA/0 to 10VDC.
- **Measuring Ranges** : -105.4 to 105.4 mm WC, -1034 to 1034 Pascals and 8 types of units equivalent values.
- **Accuracy** : $\pm 1.5\%$ for all units.
- **Resolution** : 0.1mm WC, 1Pascal s.
- **Power supply** : 24VDC 80mA.
- 3-different posts are available, users can choose any two ports to measure differential pressure.

Temperature and RH Transmitter

The NIA "Temperature and RH Transmitter" has been specially developed keeping process measurement requirements of CLEAN ROOMS in Pharmaceutical industries. RHT105 is a high-performance 4-wired Relative Humidity and Temperature Transmitter. Its relative humidity and temperature transmitter is available in surface mount, duct mount and remote sensor versions. This transmitter is accurate to 1.5% for relative humidity and within 0.3°C for temperature. RHT105 transmits two isolated output signals to other devices such as a programmable Logic Controller (PLC) or Solid-Static Relay (SSR). Signal isolation protects your devices and machines downstream of our instruments from damaging fluctuations in the input signal. The RH+T transmitter is user programmable by a PC, connected via Configuration software by NIA.

Specifications

Input Sensor Type:

RH : Capacitive sensor.
Temperature : Semiconductor (Band gap) sensor.

Output Type:

Output1 : 4-20mA/0-10VDC.
Output2 : 4-20mA/0-10VDC.

Measuring Range:

RH : 0-100% Non-condensing.
Temperature : 0 to 50°C.

Accuracy:

- 0.3°C for Temperature.
- 1.5% RH for 10-90% and 2% for remaining.

Operating Voltage : 18-24VDC.

Enclosure : ABS ip65.

Sensor Tube : Ss304.

Filter : Sintered bronze.

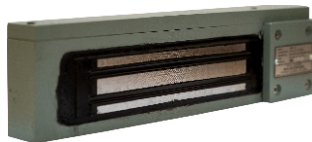
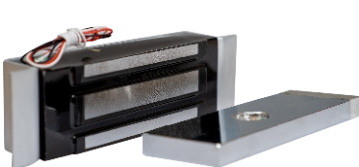
Protection : IP65 except sensor tube.

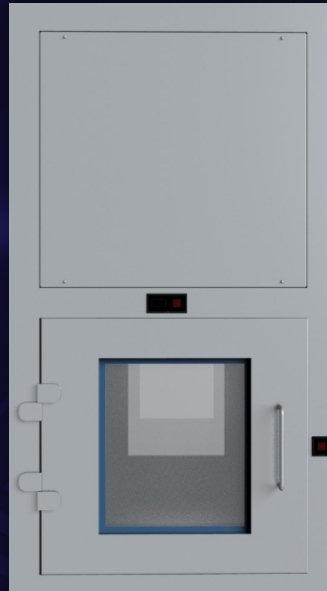
Features:

- Advanced Microcontroller based process measurement.
- Software programmable measurement range settings.
- 5-point calibration for temperature.
- 5-point calibration for RH.
- High input impedance.
- Software calibration - No trim pots.
- PC programmable.
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Accessories





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